

Purpose

This R script analyzes a small dataset to explore the relationship between the number of hours students study and their exam scores.

The script calculates average study hours, average exam scores, and the correlation between study hours and exam scores. It also creates a scatter plot to visually display the relationship.

Inputs

The script uses a small dataset containing:

- * Student
- * Study hours
- * Exam score

The data is created directly in the R script, so no external data file is required.

Outputs

The script produces:

- * The student dataset
- * The average number of study hours
- * The average exam score
- * The correlation between study hours and exam scores
- * A scatter plot showing study hours and exam scores

R Code

The following code creates the dataset:

```
```{r}
students <- data.frame(
 Student = c("A", "B", "C", "D", "E"),
 Study_Hours = c(2, 3, 4, 5, 6),
 Exam_Score = c(58, 65, 72, 81, 88)
)

print(students)
```
```

Calculating the Averages

The following code calculates the average study hours and average exam score:

```
```{r}
average_study_hours <- mean(students$Study_Hours)
print(average_study_hours)

average_exam_score <- mean(students$Exam_Score)
print(average_exam_score)
```
```

Calculating the Correlation

The correlation measures the relationship between study hours and exam scores.

```
```{r}
correlation <- cor(
 students$Study_Hours,
 students$Exam_Score
)
```

```
print(correlation)
```\n
```

Creating the Plot

A scatter plot is used to display the relationship between study hours and exam scores.

```
```\n{r}\nplot(\n  students$Study_Hours,\n  students$Exam_Score,\n  main = "Study Hours and Exam Scores",\n  xlab = "Study Hours",\n  ylab = "Exam Score"\n)\n```\n
```

### ## Summary

This script demonstrates a basic data analysis workflow in R. It creates data, performs simple calculations, examines the relationship between two variables, and produces a visualization.